How many ways are there to put *n* distinct objects into *k* distinct boxes such that every box contains at least one object?

Please explain elaborately,as i am highly confused in this part.

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6 ANSWERS

Raziman Thottungal Valapu, Combinatorics enthusiast
2 upvotes by Quora User and Nenad Cakic.

Stirling numbers of the second kind count the number of ways of splitting n objects into k non-empty groups. Multiplying this quantity by k! gives you the answer you want.

Stirling numbers are not so easy as binomial coefficients, but the above linked Wikipedia article contains pretty much everything you need to know.

Written 2 Apr, 2014. 1,259 views. Asked to answer by Anonymous.

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Baljeet Singh, Number Games!
1 upvote by Nenad Cakic.

This question is equivalent to asking numbers of ways to place n-non identical balls in m non-identical boxes, where no boxes is empty.

Suppose we have four different English alphabets, 1 a, 1 b , 1 c, 1 d. We have to put it into two identical boxes so in how different many ways we can do it?

	1	2	3	4	5	6	7
box1	a	b	c	d	ad	bc	ab
box2	bcd	acd	abd	abc	bc	ad	cd

So total is 7.

Let S(n,m) be the no of ways to put n non-identical boxes into m identical boxes with no empty boxes allowed.

S(n,0) = 0 (n different balls in 0 box in 0 way)
S(n,1) = 1 (n balls in 1 box has only one i.e. all in one)
S(n,n) = 1 (n balls in n identical boxes in 1 way(no empty boxes))
S(n,2) = ?, S(4,2) = 7(we have found above).

Suppose we have n balls and two boxes. All the balls are numbered as b1, b2, b3, b4, b5, ... bn.

Let we put first ball b1 in one of two boxes as they are identical so it does not matter much. Now we have n - 1 balls left, so we have 2 choices for each of these n - 1 balls either it will be in same box or in other one. Therefore, there are 2^{n-1} ways. But if all the balls came in one box then one box become empty so we have to subtract **1** by considering this situation.

So, **S(n,2)** = $2^{n-1} - 1$ ways to put n different balls in 2 identical box with no empty box allowed.

So, S(4,2) = $2^{4-1} - 1$ => $8 - 1$ = 7

So in general we will find it in the same way-

We have n different balls b1, b2, b3, b4, ..., bn and m identical boxes with no empty boxes allowed. So now we pick one ball let suppose its b1. We get two cases -

1. We place b1 in one of the boxes on its own, so for remaining n - 1 balls we have m - 1 boxes left which is equivalent to **S(n - 1, m - 1)**
2. b1 is not alone in its box. So therefore, we have other n - 1 balls left for m boxes such that no empty boxes allowed which is equivalent to **S(n - 1,m).** But we have consider that b1 can be put in any of these m boxes in **m ways** so therefore **m*S(n - 1,m).**

So we have final result as
S(n,m) = S(n - 1,m - 1) + m*S(n - 1,m)

So the given question is equivalent to **S(n,m)*m! which is equivalent to permutation of picking n from m non-identical element with repetition allowed.**

We can prove it by using exponential generating function.

$$G_e(x) = (x + \frac{x^2}{2!} + \frac{x^3}{3!} + ...)^m$$

as empty element not allowed so 1 is not in generating function.

$$(e^x - 1)^m = \sum_{h=0}^{m} \binom{m}{h} (-1)^h e^{(m-h)x}$$
$$(e^x - 1)^m = \sum_{0}^{\infty} \frac{(m-h)^n}{n!} x^n$$

$$e^{(m-h)x} = 1 + \frac{(m-h)}{1!} x + \frac{(m-h)^2}{2!} x^2 + ...$$

$$G_e(x) = \sum_{h=0}^{m} \binom{m}{h} (-1)^h \sum_{n=0}^{\infty} \frac{(m-h)^n}{n!} x^n$$

$$G_e(x) = \sum_{n=0}^{infty} \frac{x^n}{n!} \sum_{h=0}^{m} \binom{m}{h} (-1)^h (m-h)^n$$

matching the coefficient with Stirling formula(Stirling numbers of the second kind),
So,
$$m!.S(n,m) = \sum_{h=0}^{m} \binom{m}{h} (-1)^h (m-h)^n$$
is the solution for n different balls into m different boxes with no empty boxes allowed

Written 29 Dec. 806 views.

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Shreyans Doshi, Sometimes I use Quora and sometimes I... (more)

Here main requirement is n>=k,

So, the formula will be :
[$nC0(n-0)^k-nC1(n-1)^k+nC2(n-2)^k+\cdots+(-1)^(n-1)nC(n-1)(1)^k$]

Distributing Balls into Boxes

Written 29 Dec. 295 views.

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Rajendra Kumar Uppal, former PhD student at IIT, Kanpur
1 upvote by Imran Al Noor.

You require, $n >= k$ for this to be true. Because if number of objects are less than number of boxes then at least one box will be empty. (Reverse pigeonhole principle?)

Now if $n >= k$ then number of ways will be equal to choosing k boxes out of n objects which is `'n choose k'`, see Binomial coefficient .

Written 1 Apr, 2014. 509 views.

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Anonymous
1 upvote by Balamurugan Avudaiappan.

I don't think the above answer makes sense!
suppose 5 balls have to be put in 3 boxes

so total ways(including when some boxes are empty)=3^5

There can be a case when 2 boxes get all balls: 3C2 (2^5-2)

(Because there are two instances in which 1 box gets all the balls and we don't want to over count as we are taking the single box case differently.)

There can be a case when 1 box gets all the box =3C1*1

Hence cases in which all boxes atleast get one ball= 3^5- 3C2 (2^5-2)-3C1*1=150

But what you are saying is definitely wrong

Written 1 Apr, 2014. 457 views.

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You have 6 identical balls and 6 (distinct) boxes numbered 1 through 6. How many ways can the 6 balls be distributed amongst the boxes?

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